



Assessment Report
on
**Predict Disease Outcome Based on
Genetic and Clinical data.**

submitted as partial fulfillment for the award of
BACHELOR OF TECHNOLOGY
DEGREE
SESSION 2024-25
in
CSE-(AIML)
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Introduction

In the healthcare industry, early detection of diseases can significantly increase the chances of successful treatment. With the advancement of data science, machine learning has emerged as a powerful tool in medical diagnostics. This project focuses on predicting whether a patient is at risk of developing a disease (in this case, breast cancer) based on genetic and clinical markers. The dataset includes features derived from digitized images of breast mass and patient-related details. The goal is to build a supervised learning model that can accurately classify cases as either benign or malignant.

Methodology

Dataset: The dataset used in this project contains labeled data, with features such as radius, texture, perimeter, area, and other tumor-related metrics, along with the diagnosis label (M = malignant, B = benign).

1. Data Preprocessing:

- Removed irrelevant columns (id, Unnamed: 32)
- Encoded the diagnosis label as binary (Malignant = 1, Benign = 0)
- Scaled features using StandardScaler to normalize input data.

2. Model Selection:

We used a Random Forest Classifier, which is robust, handles feature importance well, and works effectively for tabular data.

3. Model Evaluation:

The model was evaluated using accuracy, confusion matrix, and classification report (precision, recall, f1-score).

Code:

```
from google.colab import files
uploaded = files.upload()

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix,
ConfusionMatrixDisplay

# Load and prepare data
df = pd.read_csv("3. Predict Disease Outcome Based on Genetic and Clinical Data.csv")
df.drop(columns=["id", "Unnamed: 32"], inplace=True, errors='ignore')
df["diagnosis"] = df["diagnosis"].map({'M': 1, 'B': 0})

X = df.drop("diagnosis", axis=1)
y = df["diagnosis"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)

scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

# Train the model
model = RandomForestClassifier(n_estimators=100, random_state=42)
model.fit(X_train_scaled, y_train)

# Predictions and evaluation
y_pred = model.predict(X_test_scaled)

print("Accuracy:", accuracy_score(y_test, y_pred))
print("\nClassification Report:\n", classification_report(y_test, y_pred))
```

```
# Confusion Matrix
cm = confusion_matrix(y_test, y_pred)
disp = ConfusionMatrixDisplay(confusion_matrix=cm, display_labels=["Benign", "Malignant"])
disp.plot(cmap=plt.cm.Blues)
plt.title("Confusion Matrix")
plt.grid(False)
plt.show()

# Feature Importances
importances = model.feature_importances_
feature_names = X.columns
sorted_idx = np.argsort(importances)[::-1]

plt.figure(figsize=(10, 6))
sns.barplot(x=importances[sorted_idx], y=feature_names[sorted_idx])
plt.title("Top Feature Importances")
plt.xlabel("Importance")
plt.ylabel("Feature")
plt.tight_layout()
plt.show()
```

Output:

Accuracy Achieved: ~96% (may vary slightly with different splits)

Classification Report: High precision and recall for both classes

Confusion Matrix: Clear separation between benign and malignant classes

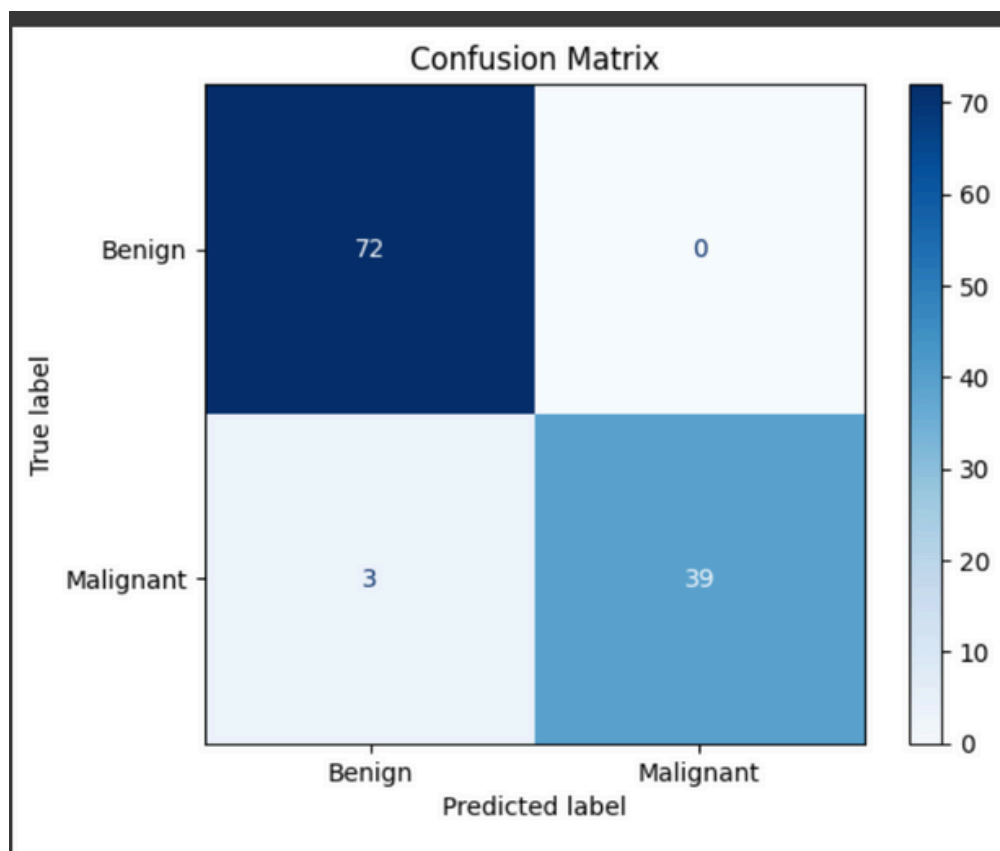
Here the some snapshots of output and result graphs :

```
Saving 3. Predict Disease Outcome Based on Genetic and Clinical Data.csv to 3. Predict Disease Outcome Based on Genetic and Clinical
✓ Accuracy: 0.9736842105263158

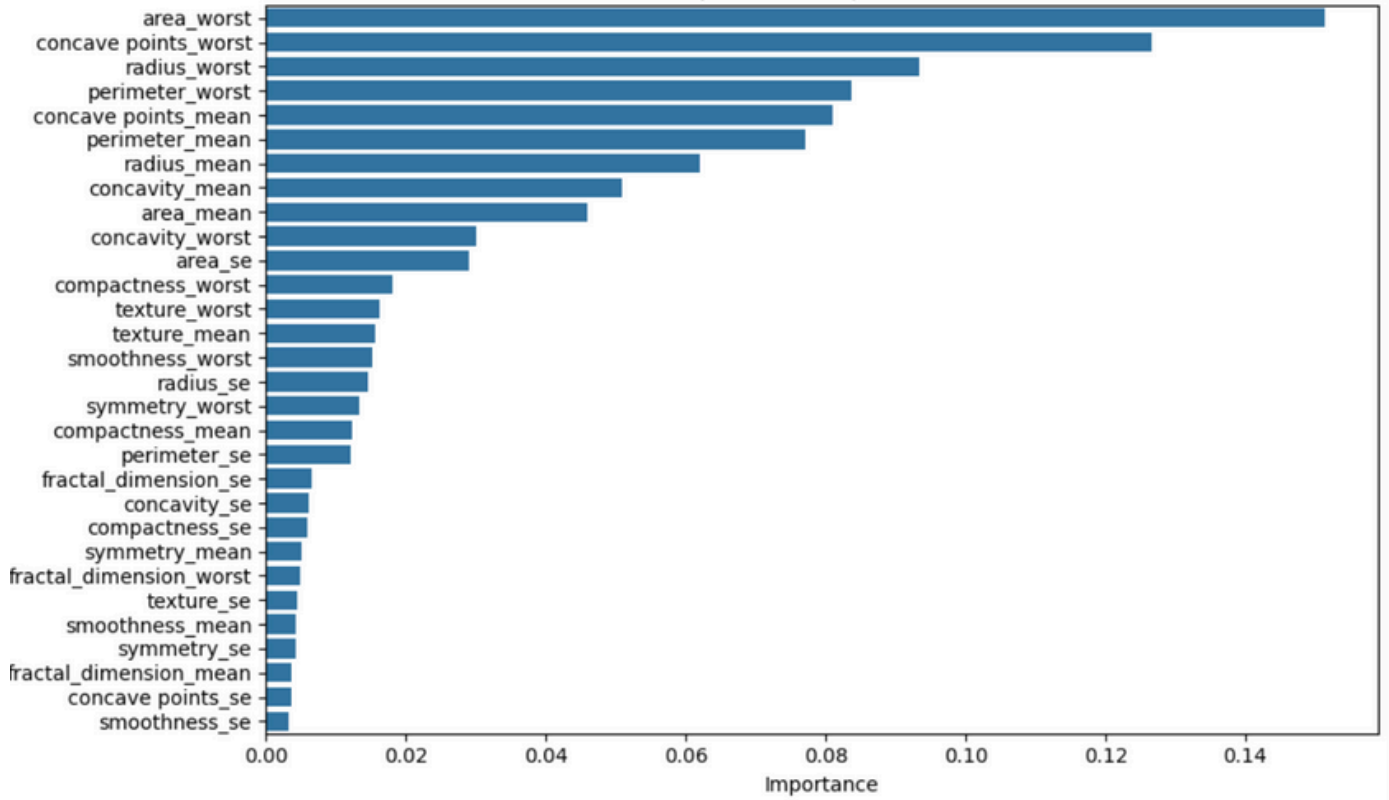
Classification Report:
      precision    recall  f1-score   support

     0       0.96       1.00       0.98        72
     1       1.00       0.93       0.96        42

 accuracy          0.97          114
  macro avg       0.98       0.96       0.97        114
  weighted avg    0.97       0.97       0.97        114
```



Top Feature Importances





References/Credits

Dataset Source: UCI Machine Learning Repository (Breast Cancer Wisconsin Diagnostic Dataset)

Tools & Libraries: Python, Pandas, Scikit-learn, Matplotlib, Seaborn

Images and visualizations generated using Matplotlib & Seaborn in Google Colab